

How Predictable Is ASML?

A Box–Jenkins audit of a single equity

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Time Series Analysis — Group Mini Project

The question we actually asked

We were asked to collect stock data, build a time series model, and forecast.

Early on, one thing became clear:

The honest deliverable is not a forecast.

It is a **verdict on forecastability**.

Why the framing matters

- A project that *promises* forecasts and delivers noise has **failed**
- A project that *asks whether forecasting is possible* and proves it is not has **succeeded**

Same numbers. Very different scientific claim.

The data

Ticker	ASML — EUV lithography monopoly
Source	Yahoo Finance via <code>yfinance</code> , adjusted
Period	2015-01-02 → 2026-08-14 (11.6 years)
Daily observations	2,921
Last close	1,844.08
CAGR	28.98%
Annualised volatility	38.01%

Daily data for stationarity, ARIMA and volatility.

Monthly (140 points) for decomposition and exponential smoothing.

Part 1 — Data, Decomposition & Stationarity

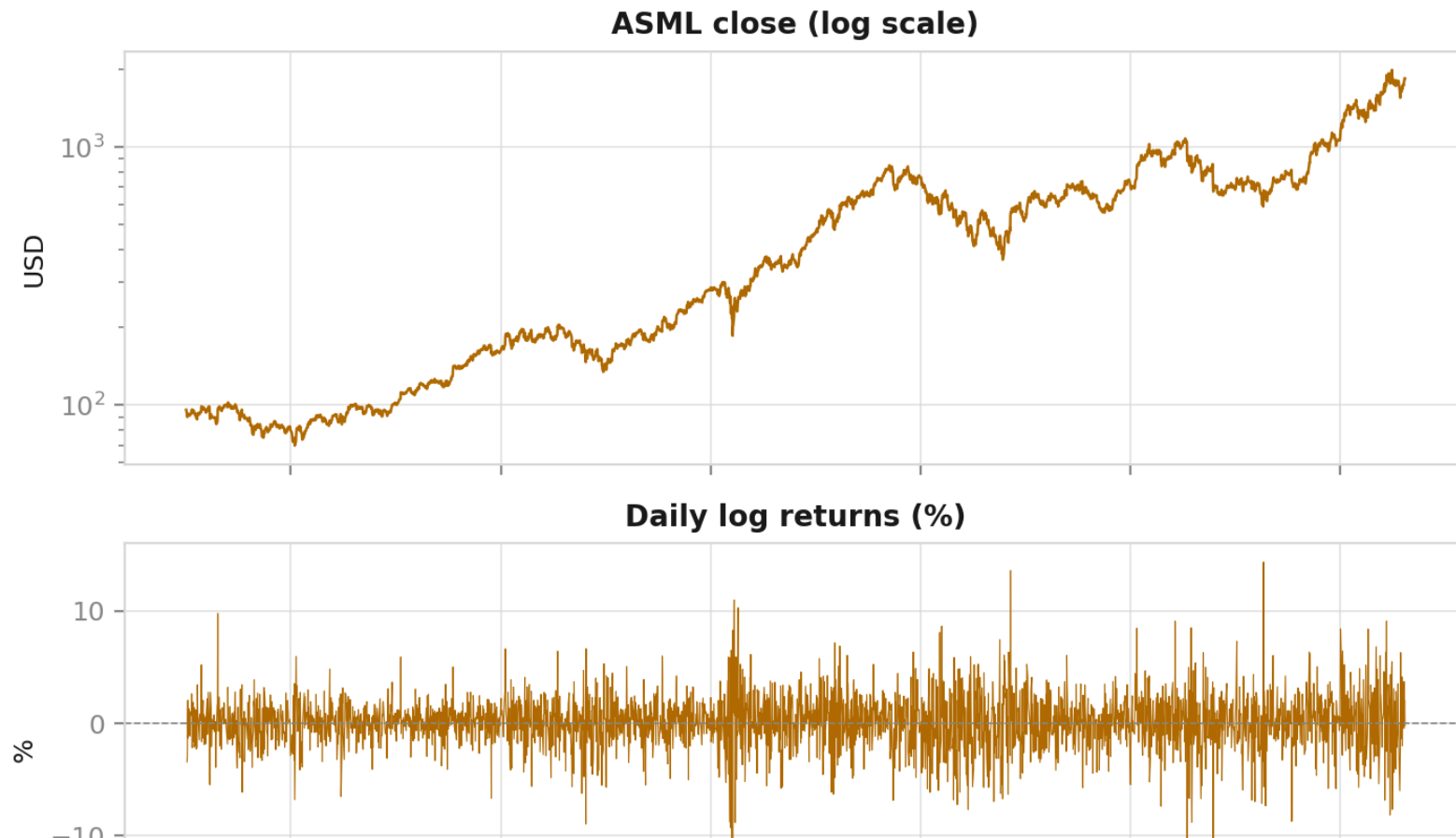
Abhinabha Das

Why we model the log price

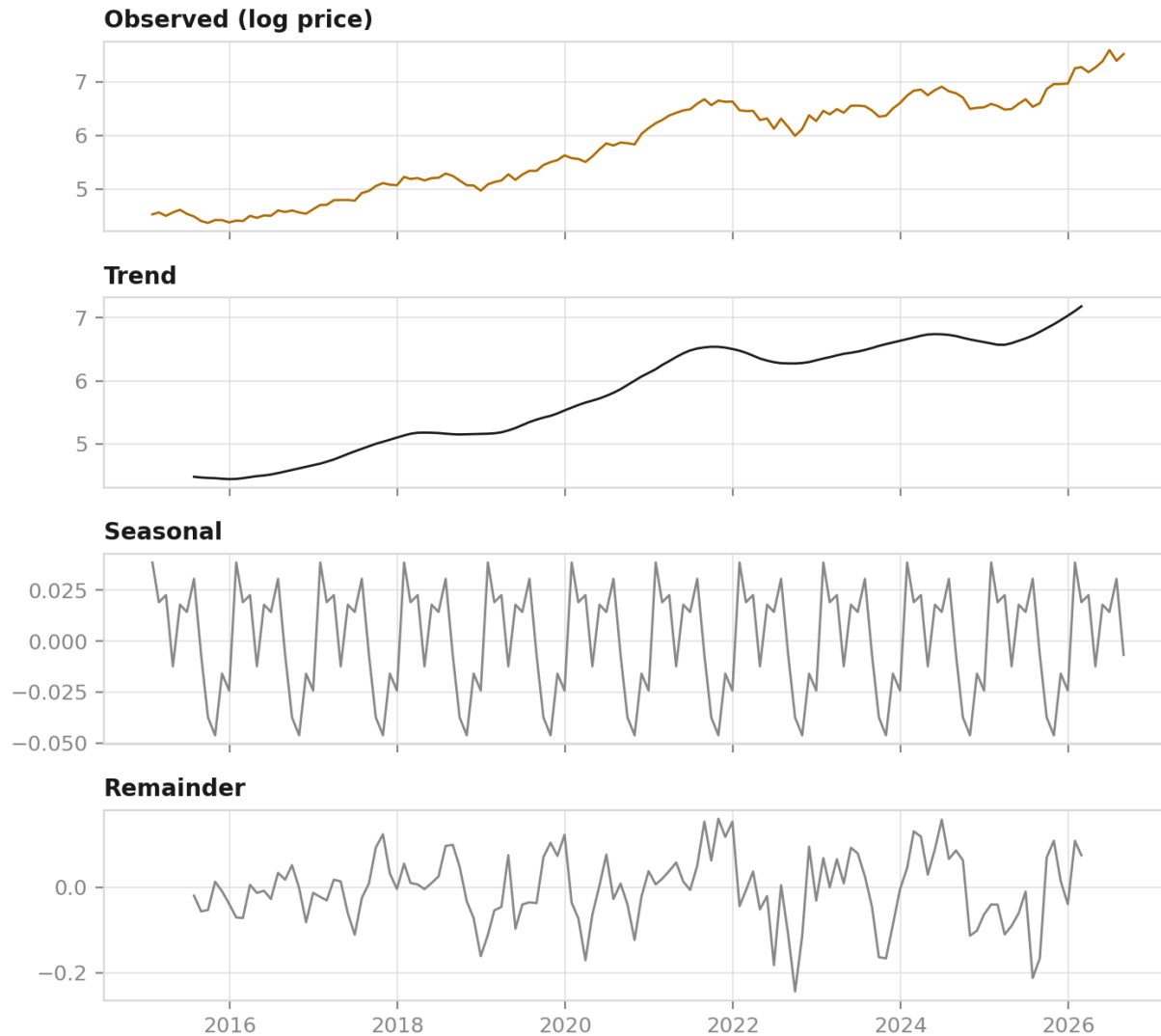
ASML compounded at **28.98% a year**.

Growth that fast is **multiplicative** — variance grows with the level.

Logs turn multiplicative growth into additive growth *and* stabilise variance.



Decomposition: trend 0.99, seasonal 0.10



There is no calendar season in a share price.

Stationarity: two tests, opposite nulls

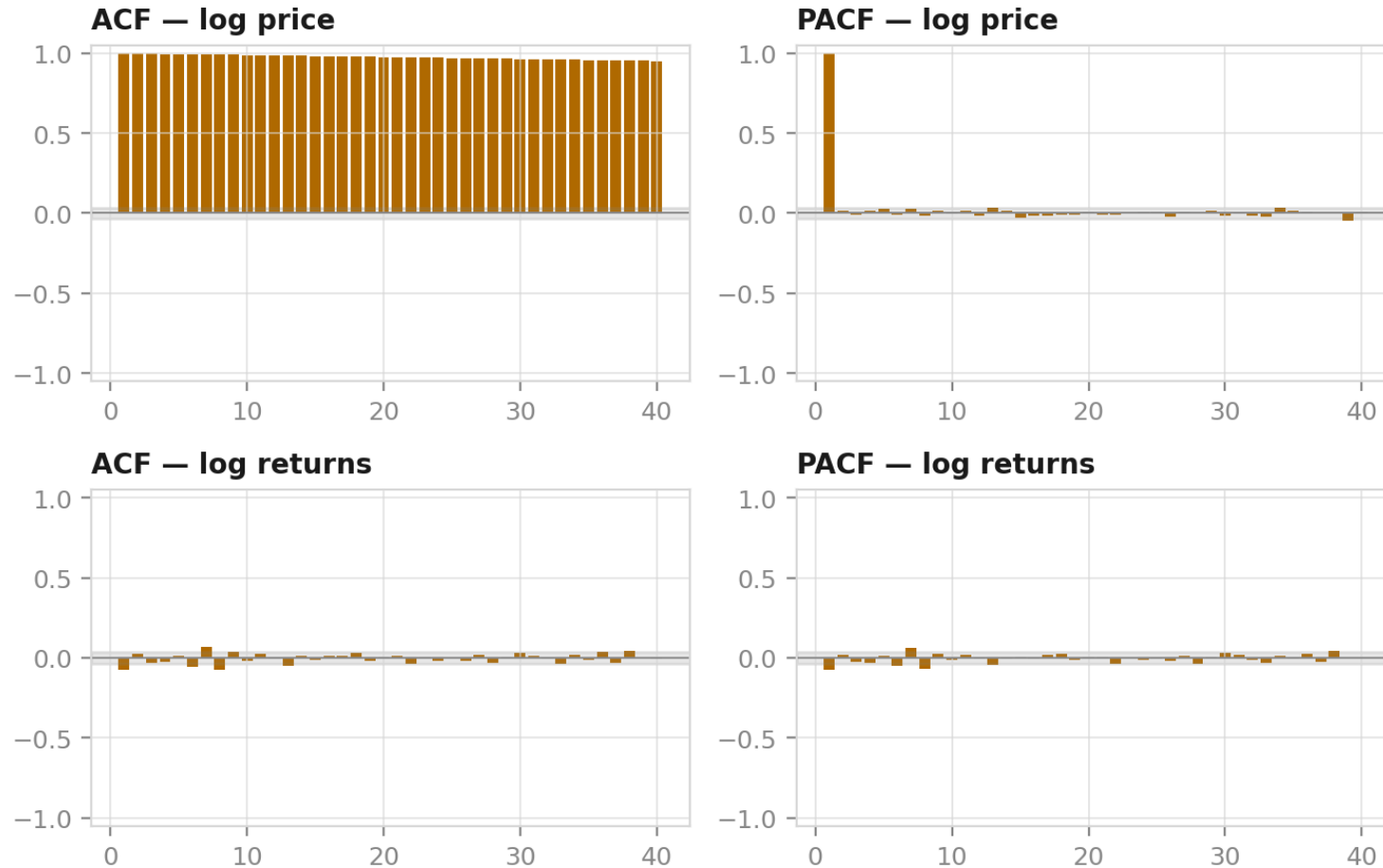
- **ADF** H_0 : unit root (non-stationary)
- **KPSS** H_0 : stationary

Series	ADF p	KPSS p	Verdict
Log price	0.9634	0.0100	Non-stationary
Δ log price	< 0.0001	0.1000	Stationary

Because the nulls are **opposed**, two-way agreement is far stronger than either test alone.

ASML log price is $I(1) \rightarrow d = 1$

Correlogram



- Log price: **all 40** lags outside the ± 0.0363 band — no fixed mean to revert to
- Returns: only **9 of 40**, none large

Part 2 — Moving Averages & Exponential Smoothing

K Suraj Das

The three-model ladder

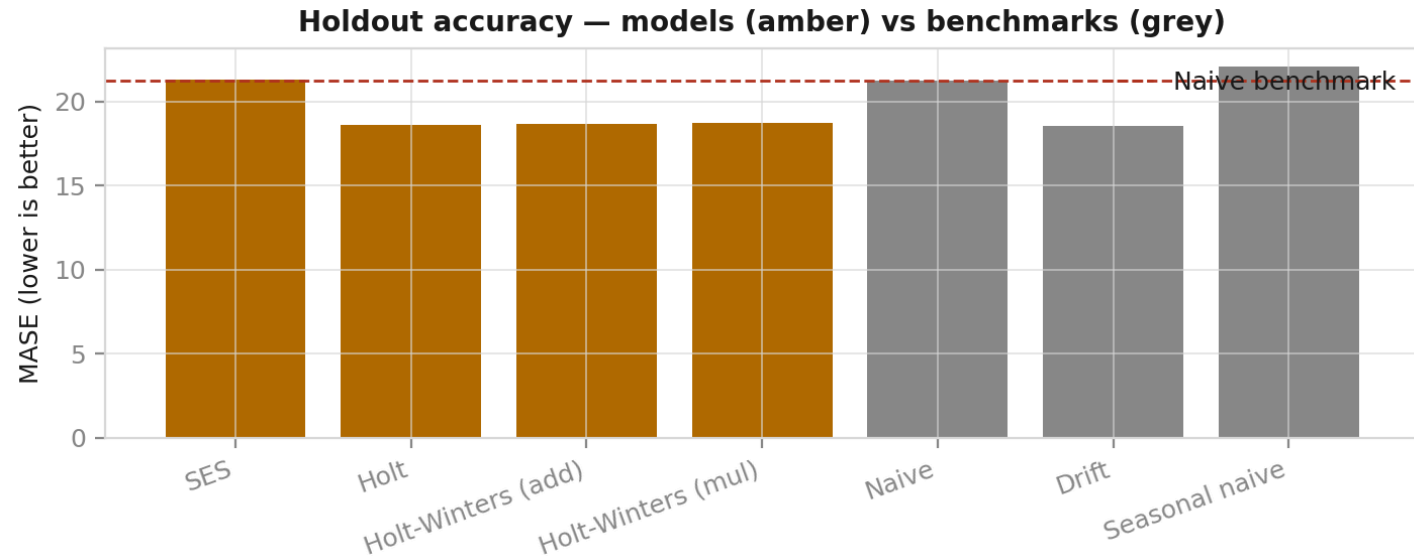
SES → level only · Holt → + trend · Holt-Winters → + season

Model	α	β	γ	MAPE
SES	0.9879	—	—	44.68%
Holt	0.9371	0.0000	—	39.19%
Holt-Winters	1.0000	0.0000	0.0000	39.60%

Three parameters tell the whole story:

- $\alpha \approx 1$ → the level is reset to whatever just happened → *a random walk in a costume*
- $\beta = 0$ → trend treated as fixed → Holt collapses onto Drift
- $\gamma = 0$ → no seasonality to learn → exactly as Part 1 predicted

Nothing beats Drift



Model	MASE		Benchmark	MASE
SES	21.279		Naive	21.248
Holt	18.625		Drift	18.565
Holt-Winters	18.653		Seasonal naive	22.081

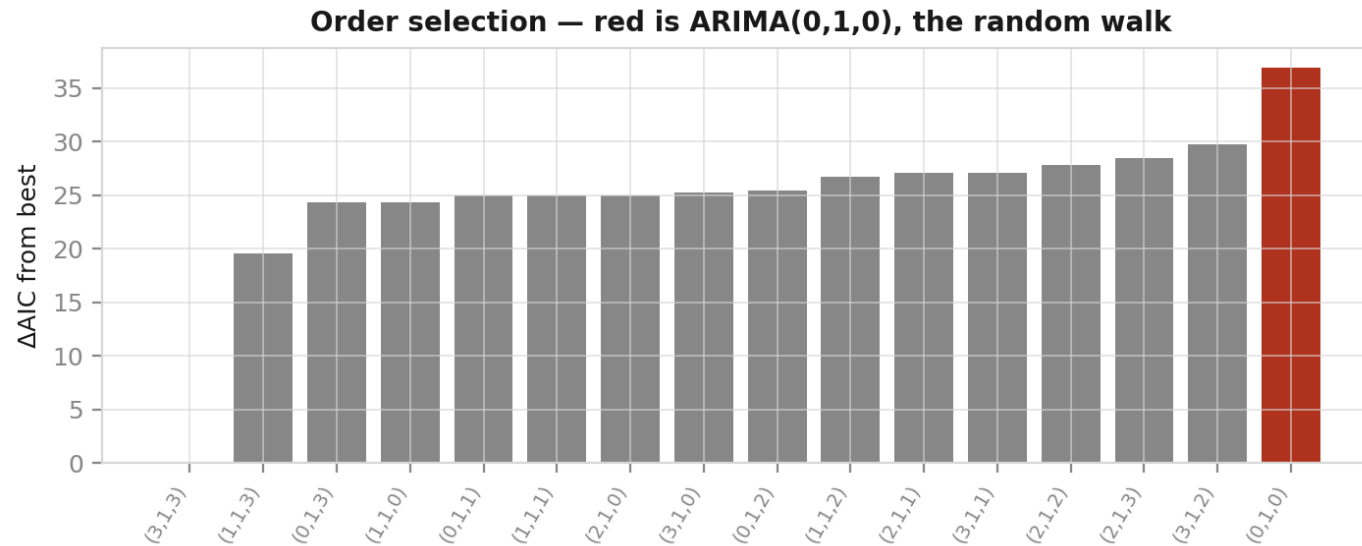
Last value + average monthly change — two parameters, no fitting — wins.

MASE divides by the one-step in-sample naive error, so values > 1 are normal at a 12-step horizon. Read the ranking, not the level.

Part 3 — ARIMA & Forecast Evaluation

Akash Kumar

Order selection: 16 candidates by AIC



Order	AIC	ΔAIC
(3,1,3)	-13,539.73	0.00
(1,1,3)	-13,520.20	19.53
(0,1,0) random walk	-13,502.80	36.93

The random walk ranks **16th of 16**. AIC *does* find real structure — this is not indecision.

But the coefficients mislead

Coefficient	Estimate
ar.L1	-0.8902
ma.L1	0.8286

All significant at 5%. Looks like strong structure.

It is not

- AR root modulus 1.1234
- MA root modulus 1.0865

They differ by ~3% → the polynomials share a **near-common factor** and nearly cancel.

A model can carry large, significant coefficients and still be a near-random walk.

Only an out-of-sample test settles it.

The trap: one residual inverts every diagnostic

`statsmodels` starts its filter from a diffuse prior. The first residual is the filter finding its footing — **not** a model error.

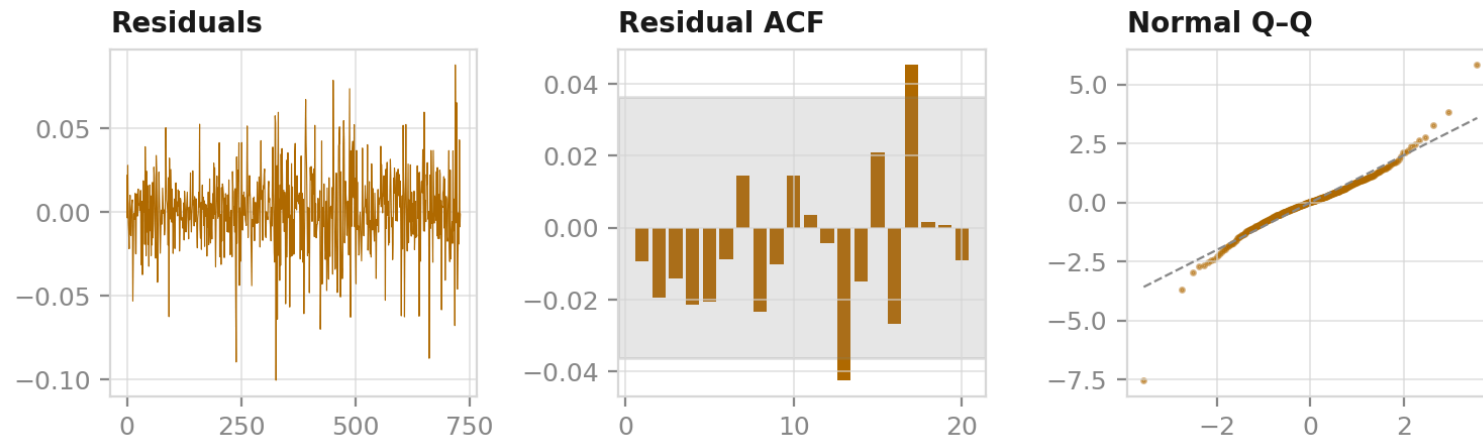
4.5638 vs $\sigma = 0.0237 \rightarrow 192.2\sigma$

	Ljung-Box p
Burn-in kept	0.99999 — "flawless white noise"
Burn-in trimmed	0.0961 — passes, but only just

The whole appearance of a perfect fit rested on one observation, and nothing in the output looks wrong.

Every diagnostic in this project routes through one shared trimming helper.

Residuals, honestly



Test	p	Meaning
Ljung-Box (10)	0.0961	no autocorrelation at 5%
Jarque-Bera	< 0.0001	not normal (kurtosis 7.2615)
ARCH-LM (10)	< 0.0001	variance is not constant

The mean model is clean — and still cannot beat naive. The other two are **findings**, and they point straight to Part 4.

The headline result

74 rolling origins, expanding window, refit every time.

Model	h=1	h=3	h=6
ARIMA(3,1,3)	7.179	13.558	24.483
Naive	7.151	13.634	24.556
Drift	7.020	13.211	23.273

Diebold–Mariano p vs naive: 0.235 → 0.888. **Not one below 0.05.**

No model beats naive at any horizon

Drift ranks first — but a ranking without a significant margin **is not a win**. Saying otherwise would be the easiest mistake in this project.

Where the model genuinely fails

Horizon	Coverage	Nominal
h = 1	82.4%	95%
h = 3	82.4%	95%
h = 6	79.7%	95%

Mean **81.5%** against 95%. The intervals are **too narrow** — the model understates risk.

Cause: ARIMA assumes one constant variance. ARCH-LM rejected that at $p < 0.0001$.

Averaging volatility across calm and turbulent regimes understates risk *exactly* when it matters.

Why a null result is a strong result

Four independent tools. Four different labs. One conclusion.

1. ADF **and** KPSS agree: $I(1)$
2. ACF of returns collapses immediately
3. ARIMA's AR and MA roots nearly cancel
4. No model beats naive out of sample (DM $p \geq 0.235$)

This is weak-form market efficiency (Fama, 1970)

If past prices held exploitable information, trading would already have removed it.

The failure mode we deliberately avoided: tuning until something wins. Given enough specifications, one always will.

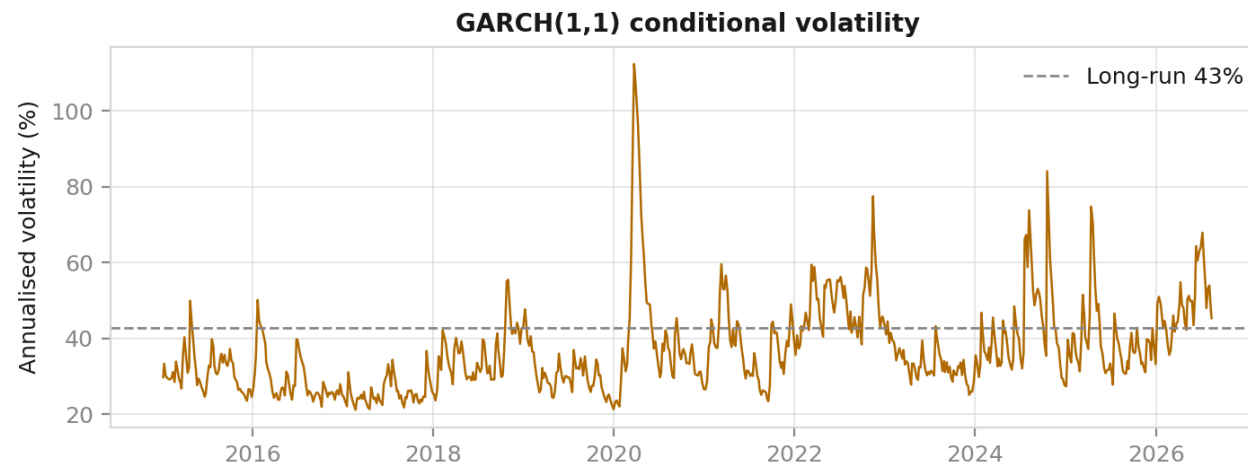
Part 4 — Volatility & Black–Scholes

Ritesh KR · beyond syllabus

The variance is forecastable

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Parameter	Value
α (shock)	0.0752
β (persistence)	0.9113
$\alpha + \beta$	0.9864
Half-life	50.7 trading days



Ljung-Box on squared standardised residuals: $p = 0.2012 \rightarrow$ ARCH fully absorbed.

The central contrast

We cannot say where ASML is going.

We can say how rough the ride will be.

- **Mean**: unforecastable — 74 origins, no significant edge over naive
- **Variance**: strongly forecastable — persistence 0.9864, half-life 50.7 days

The same data supports both claims. They are not in tension: returns are unpredictable in *direction*, highly structured in *magnitude*.

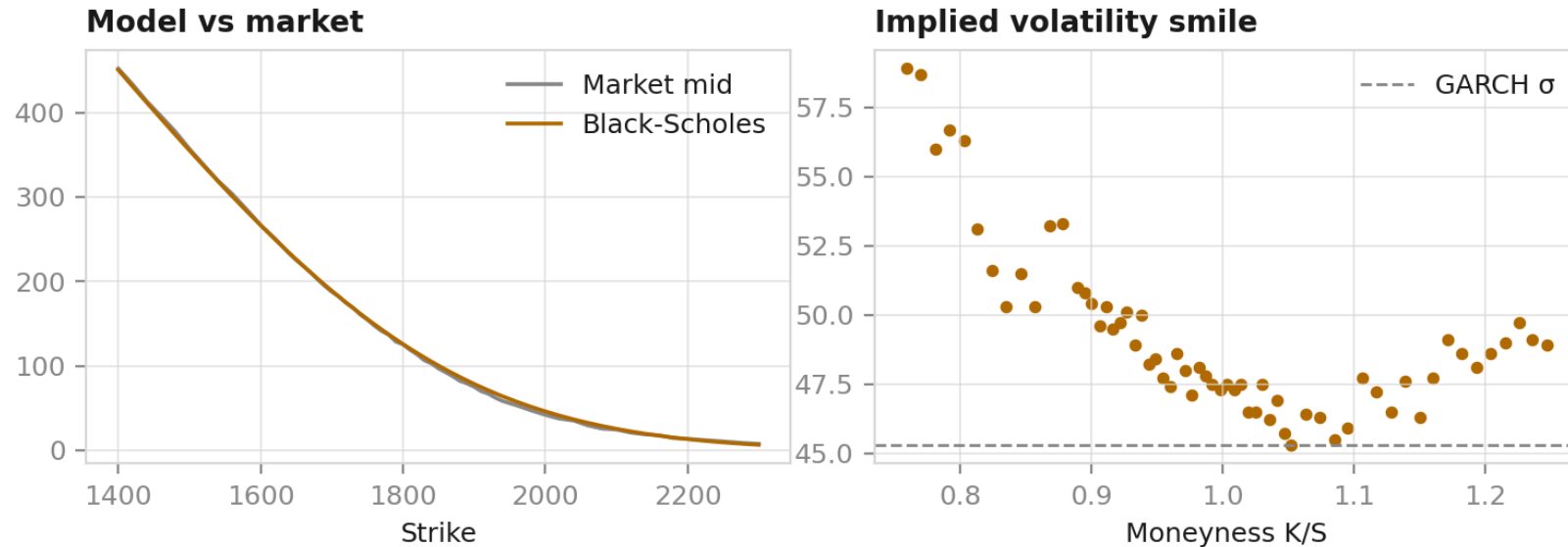
Black–Scholes against a live market

61 near-the-money calls, expiry **2026-09-18**, $r = 3.697\%$

Spot	1,844.08
ATM strike	1,840
Market mid	102.85
Black–Scholes @ GARCH σ	105.09 (+2.2%)
Market implied vol	49.2%
GARCH vol	45.3%
Gap	+3.9 points

The market charges **more** volatility than history justifies → variance risk premium.

The smile: Black–Scholes' second failure



Black–Scholes assumes **one volatility for every strike**. Implied volatility visibly **curves**.

Both departures trace to the same wrong assumption — constant Gaussian volatility — that Part 3's ARCH test and kurtosis of 7.58 had already rejected.

Part 5 — Macro-Factor Lag Analysis

Aditya Mehta · beyond syllabus

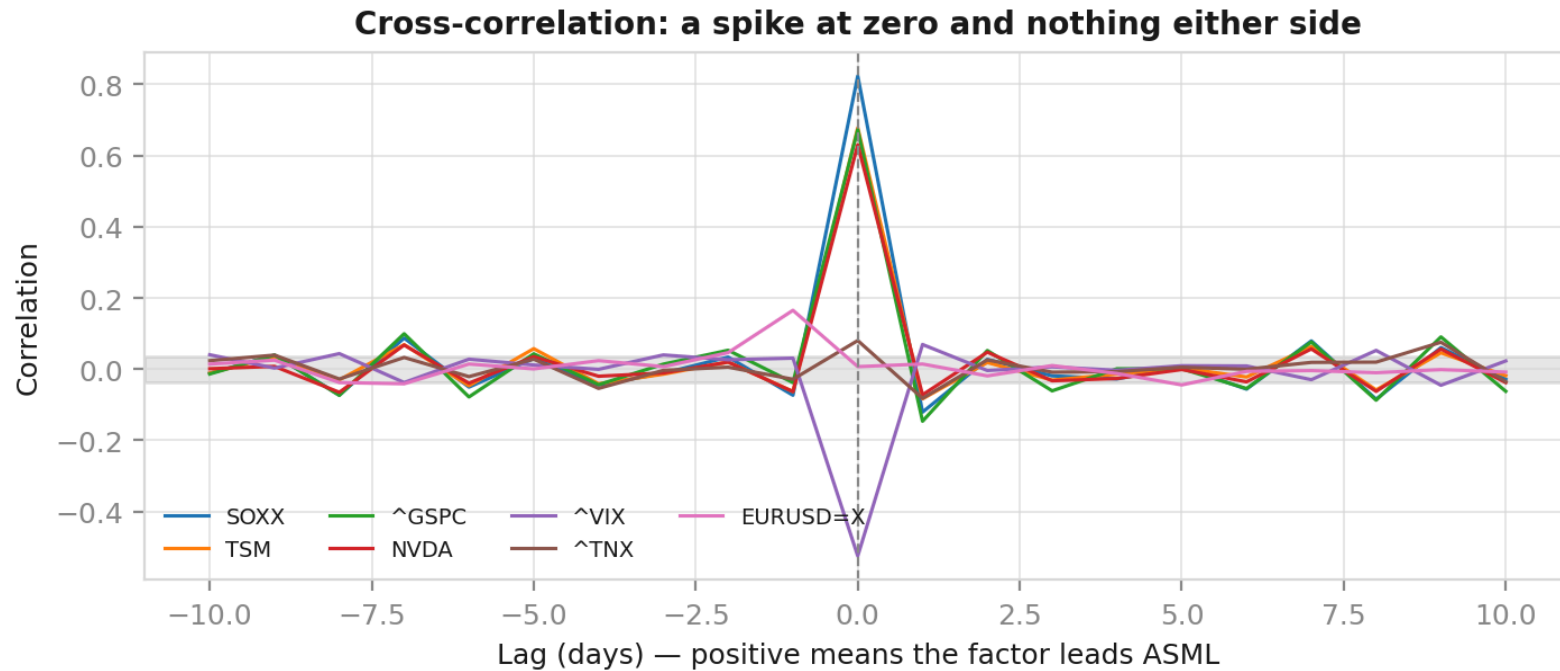
If ASML can't predict itself, can anything else?

Seven factors, 2,915 common trading days.

Factor	Same day	Best lead (+1)	Ratio
SOXX	0.8237	-0.1210	0.15
TSM	0.6780	-0.0836	0.12
^GSPC	0.6728	-0.1466	0.22
^VIX	-0.5253	0.0690	0.13

A correlation of **0.82** looks like an opportunity — until you notice *when* it exists.

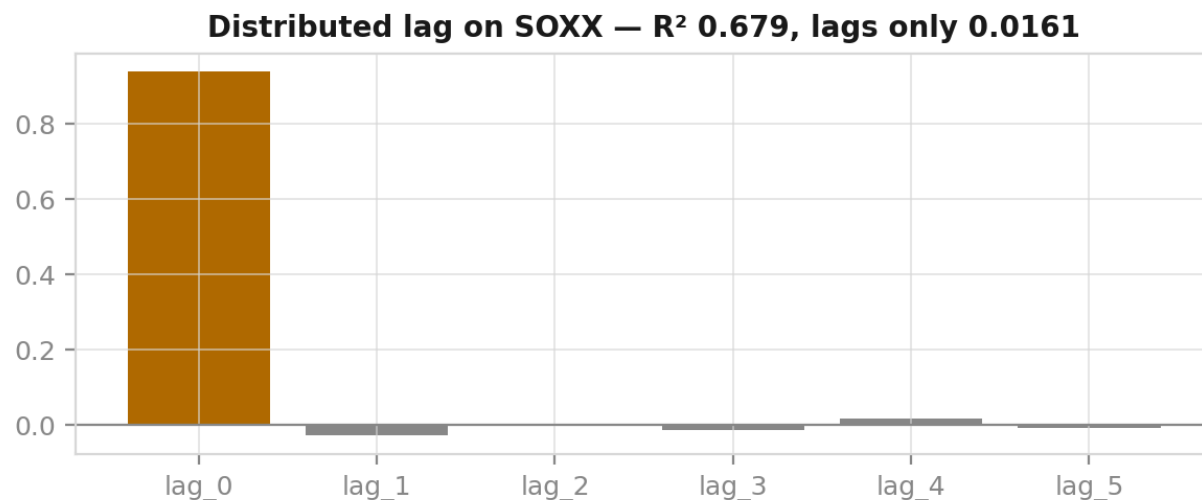
A spike at zero, and nothing either side



ASML moves *with* the semiconductor complex, not *after* it.

Note the lag-1 values are **negative** — the opposite sign to lag 0. That is reversal and bid-ask bounce, not prediction. A true leader would keep the same sign.

Quantifying it: one variable carries everything



Specification	R ²
With the same-day term	0.6795
Lags only — what a forecaster could use	0.0161

Removing the one variable you can't know in advance destroys 97.6% of the explanatory power

High correlation and zero predictability coexist comfortably.

Granger: reporting the awkward result

Lags 1–5, Bonferroni threshold $\alpha = 0.01$.

Four factors are significant: SOXX, TSM, ^GSPC, ^TNX.

This appears to contradict the section. It does not:

1. **Significance $\neq$ usefulness.** At $n = 2,915$ a test detects effects explaining $< 1\%$ of variance — precisely what the R^2 above measures.
2. **Part 3 already showed** what happens to small in-sample effects out of sample.

Granger causality is a claim about **forecast improvement**, not cause. For a chip-equipment maker and a semiconductor index, a **common driver** is by far the likeliest explanation.

Conclusion

What we found

The direction of ASML is not forecastable

- $I(1)$; near-cancelling ARIMA roots; no model beats naive across 74 origins (DM $p \geq 0.235$)
- Four independent confirmations = weak-form market efficiency (Fama, 1970)

The magnitude of its moves is

- GARCH persistence 0.9864, half-life 50.7 days, ARCH fully absorbed

And we found where our own model fails

- Prediction intervals cover 81.5%, not 95% — the constant-variance assumption breaking

Limitations & further work

- Single asset, single regime (2015–2026 was one extraordinary bull run)
- Linear models only — non-linear or regime-switching structure would be invisible
- **SARIMA not fitted** — seasonal strength 0.098 did not justify it; the natural next step
- Daily closes hide microstructure; bid-ask bounce likely explains the negative lag-1 values
- No transaction costs modelled

Reproducibility: every figure and number comes from `make_figures.py`, which imports the same modules that serve the dashboard. One implementation — a slide cannot disagree with the app.

Thank you

Questions?

Live dashboard · 6 sections · FastAPI + React